

List of Scientific Publications

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The following publications were produced during the period of this doctoral research (2020–2026). Publications marked with (★) are directly related to the content of the dissertation; for these, the author was primarily responsible for the conceptual design, implementation, experiment design, evaluation, and writing.

Publications Related to the Dissertation

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- [1] **C. Gerhardt** and W. Broll, “SkyCloud360: Sky and Cloud Segmentation in Equirectangular Images,” in *Proc. 10th Int. Conf. Image, Vision and Computing (ICIVC)*, Chengdu, China, 2025, pp. 48–58. DOI: [10.1109/ICIVC66358.2025.11200399](https://doi.org/10.1109/ICIVC66358.2025.11200399). (Chapter 8)
 - [2] **C. Gerhardt**, F. Weidner, and W. Broll, “SkyCloud: Neural Network-Based Sky and Cloud Segmentation from Natural Images,” in *Proc. 8th Int. Conf. Image, Vision and Computing (ICIVC)*, Dalian, China, 2023. DOI: [10.1109/ICIVC58118.2023.10270450](https://doi.org/10.1109/ICIVC58118.2023.10270450). (Chapter 7)
 - [3] **C. Gerhardt**, F. Weidner, and W. Broll, “OUTSIDE: Multi-Scale Semantic Segmentation of Universal Outdoor Scenes,” in *Proc. IEEE 23rd Int. Workshop Multimedia Signal Processing (MMSP)*, Tampere, Finland, 2021, pp. 1–6. DOI: [10.1109/MMSP53017.2021.9733652](https://doi.org/10.1109/MMSP53017.2021.9733652). (Chapter 6)
 - [4] **C. Gerhardt** and W. Broll, “Neural Network-Based Traffic Sign Recognition in 360° Images for Semi-Automatic Road Maintenance Inventory,” in *Proc. IEEE 23rd Int. Conf. Intelligent Transportation Systems (ITSC)*, 2020, art. no. 9294610. DOI: [10.1109/ITSC45102.2020.9294610](https://doi.org/10.1109/ITSC45102.2020.9294610). (Chapter 5)
 - [5] **C. Gerhardt**, M. M. Salman, and W. Broll, “AWARE: Augmentation for Weather-Adverse Robustness Evaluation — Benchmarking Strategies for Improved Semantic Segmentation under Challenging Outdoor Conditions,” *IEEE Access*, 2025. (Under review.) (Chapter 9)

Further Publications During the Doctoral Period

- [6] S. Arévalo Arboleda, J. Hartbrich, F. Weidner, S.B. Fishedick, **C. Gerhardt**, K. Richter, C. Kunert, B. Vorhof, H.-M. Groß, W. Broll, and A. Raake, “Robot, Avatar, or Human: The Impact of Partner Representation and Task on the Communication Experience,” *Proc. ACM Hum.-Comput. Interact.*, vol. 9, no. CSCW, art. CSCW472, 2025. DOI: [10.1145/3757653](https://doi.org/10.1145/3757653).
- [7] E. Makled, **C. Gerhardt**, T. Schwandt, F. Weidner, and W. Broll, “Evaluating Behavioral Realism in AR and VR: A Comparison of Single-Point IK and Full-Body Motion Capture Virtual Humans,” *The Visual Computer*, vol. 41, no. 14, pp. 11721–11733, 2025. DOI: [10.1007/s00371-025-03934-5](https://doi.org/10.1007/s00371-025-03934-5).
- [8] V. Mikhailova, C. Kunert, J. Hartbrich, T. Schwandt, **C. Gerhardt**, A. Raake, W. Broll, and N. Döring, “Addressing the Avatar in the Room: A User Study on Older Adults’ Experiences with a Wearable Augmented Reality Communication System,” in *Human Aspects of IT for the Aged Population: ITAP 2025, HCII 2025, Proceedings, Part I*, LNCS 15809, Springer, 2025. DOI: [10.1007/978-3-031-92707-2_21](https://doi.org/10.1007/978-3-031-92707-2_21).
- [9] V. Mikhailova, C. Kunert, J. Hartbrich, T. Schwandt, **C. Gerhardt**, A. Raake, W. Broll, and N. Döring, “Work-in-Progress: Older Adults’ Experiences with an Augmented Reality Communication System,” in *Proc. ACM Int. Conf. Interactive Media Experiences (IMX)*, Stockholm, Sweden, 2024. DOI: [10.1145/3639701.3663641](https://doi.org/10.1145/3639701.3663641).
- [10] E. Makled, **C. Gerhardt**, T. Schwandt, F. Weidner, and W. Broll, “Investigating Behavioral Realism of Single-Point IK Animated Avatars of Others in AR and VR,” in *Proc. Int. Conf. Cyberworlds (CW)*, Kofu, Japan, 2024, pp. 1–8. DOI: [10.1109/CW64301.2024.00010](https://doi.org/10.1109/CW64301.2024.00010).
- [11] F. Weidner, J. Hartbrich, S. Arévalo Arboleda, C. Kunert, C. Schneiderwind, C. Diao, **C. Gerhardt**, T. Surdu, W. Broll, S. Werner, and A. Raake, “Eyes on the Narrative: Exploring the Impact of Visual Realism and Audio Presentation on Gaze Behavior in AR Storytelling,” in *Proc. ACM Symp. Eye Tracking Research & Applications (ETRA)*, Glasgow, UK, 2024. DOI: [10.1145/3649902.3653344](https://doi.org/10.1145/3649902.3653344).
- [12] V. Mikhailova, **C. Gerhardt**, C. Kunert, T. Schwandt, F. Weidner, W. Broll, and N. Döring, “Age and Realism of Avatars in Simulated Augmented Reality: Experimental Evaluation of Anticipated User Experience,” in *Proc. IEEE Conf. Virtual Reality and 3D User Interfaces (VR)*, Orlando, FL, 2024, pp. 83–93. DOI: [10.1109/VR58804.2024.00032](https://doi.org/10.1109/VR58804.2024.00032).

- [13] S. Arévalo Arboleda, C. Kunert, J. Hartbrich, C. Schneiderwind, C. Diao, **C. Gerhardt**, T. Surdu, F. Weidner, W. Broll, S. Werner, and A. Raake, “Beyond Looks: A Study on Agent Movement and Audiovisual Spatial Coherence in Augmented Reality,” in *Proc. IEEE Conf. Virtual Reality and 3D User Interfaces (VR)*, Orlando, FL, 2024, pp. 502–512. DOI: [10.1109/VR58804.2024.00071](https://doi.org/10.1109/VR58804.2024.00071).
- [14] F. Weidner, G. Böttcher, S. Arévalo Arboleda, C. Diao, L. Sinani, C. Kunert, **C. Gerhardt**, W. Broll, and A. Raake, “A Systematic Review on the Visualization of Avatars and Agents in AR & VR Displayed Using Head-Mounted Displays,” *IEEE Trans. Vis. Comput. Graphics*, vol. 29, no. 5, pp. 2596–2606, 2023. DOI: [10.1109/TVCG.2023.3247072](https://doi.org/10.1109/TVCG.2023.3247072).

Pre-Doctoral Publications

- [15] **C. Gerhardt**, P. Aichroth, and S. Mann, “Selective Face Encryption in H.264 Encoded Videos,” in *Proc. IEEE Visual Communications and Image Processing (VCIP)*, 2017, pp. 1–4. DOI: [10.1109/VCIP.2017.8305040](https://doi.org/10.1109/VCIP.2017.8305040).